

7. The Normal distribution

Recall, $F(x)$ is the distribution function of X if $F(x) = P(X \leq x)$.

X is absolutely continuous if $\exists f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$F(x) = \int_{-\infty}^x f(x) dx$$

We call $f(x)$ the density function of X .

Prop. Let X be an absolutely continuous random variable with density f_X . Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function with nowhere 0 derivative. If $Y = g(X)$, then the density function of Y is

$$f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|}$$

proof, Since the derivative of g is nowhere 0, g is either strictly monotonically increasing or strictly monotonically decreasing.

I/ g is increasing, g' always positive

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y)) \\ \Rightarrow f_Y(y) &= \frac{dF_Y(y)}{dy} = \underbrace{F_X'(g^{-1}(y)) \cdot (g^{-1})'(y)}_{\text{chain rule}} = \frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))} = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|} \end{aligned}$$

g is strictly increasing

II/ g is decreasing, g' always negative

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) = P(X > g^{-1}(y)) = 1 - F_X(g^{-1}(y)) \\ \Rightarrow f_Y(y) &= \frac{dF_Y(y)}{dy} = - \frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))} = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|} \end{aligned}$$

g is strictly decreasing $\square$

I. Normal distribution

def. We say that X is a normal random variable / gaussian random variable with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$ if its density function is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Notation, $X \sim N(\mu, \sigma^2)$

Prop: $\varphi(x)$ is indeed a density function.

proof: $\forall x \in \mathbb{R}, \varphi(x) > 0$ so we only need to prove $\int_{-\infty}^{\infty} \varphi(x) dx = 1$.

$$I_1 = \int_{-\infty}^{\infty} \varphi(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2}} dy > 0$$

$$y = \frac{x-\mu}{\sigma} \Rightarrow dx = \sigma dy$$

$$I_2 = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}} dx dy = (*)$$

Switch to polar coordinates: $x = r \cos \alpha$, $y = r \sin \alpha$ $|Jac| = \begin{vmatrix} \frac{dx}{dr} & \frac{dx}{d\alpha} \\ \frac{dy}{dr} & \frac{dy}{d\alpha} \end{vmatrix} = \begin{vmatrix} \cos \alpha & -r \sin \alpha \\ \sin \alpha & r \cos \alpha \end{vmatrix} = r$

$$(*) = \int_0^{\infty} \int_0^{2\pi} \frac{1}{2\pi} e^{-\frac{r^2}{2}} r d\alpha dr = \int_0^{\infty} e^{-\frac{r^2}{2}} r dr = \left[-e^{-\frac{r^2}{2}} \right]_0^{\infty} = 0 - (-e^{-\frac{0^2}{2}}) = 1$$

□

def: We say that X has standard normal distribution if $X \sim N(0, 1)$.

Then X has density $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ and distribution function $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$

Prop: $\forall x > 0, \Phi(-x) = 1 - \Phi(x)$, where $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$.

proof: Since $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ is an even function we have

$$\Phi(-x) = \int_{-\infty}^{-x} \varphi(y) dy = \int_x^{\infty} \varphi(y) dy = 1 - \int_{-\infty}^x \varphi(y) dy = 1 - \Phi(x).$$

φ is even

φ is a density

□

Prop: If $X \sim N(\mu, \sigma^2)$ and $Y = \alpha X + \beta$, then $Y \sim N(\alpha\mu + \beta, \alpha^2\sigma^2)$

proof:

$$P(Y < y) = P(\alpha X + \beta < y) = P(X < \frac{y-\beta}{\alpha}) = \int_{-\infty}^{\frac{y-\beta}{\alpha}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx =$$

wlog. $\alpha > 0$

$$= \int_{-\infty}^y \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\frac{z-\beta}{\alpha}-\mu)^2}{2\sigma^2}} \frac{1}{\alpha} dz = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}\sigma\alpha} e^{-\frac{(z-(\alpha\mu+\beta))^2}{2\sigma^2\alpha^2}} dz$$

$$z = \alpha X + \beta \Rightarrow dx = \frac{1}{\alpha} dz$$

□

Cor: If $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma} \sim N(0, 1)$ and $F_X(y) = \Phi\left(\frac{y-\mu}{\sigma}\right)$

Prop. Let $X \sim N(\mu, \sigma^2)$, then $E[X] = \mu$ and $D^2[X] = \sigma^2$.

proof. Let $Y \sim N(0, 1)$. According to the previous proposition $X = \sigma Y + \mu$.
So it's enough to prove the assertion for Y .

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx = \underbrace{\int_{-\infty}^0 \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx}_{=0} + \underbrace{\int_0^{\infty} \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx}_{=0} = 0 \\ &= \left[\frac{1}{\sqrt{2\pi}} (-e^{-\frac{x^2}{2}}) \right]_0^{\infty} = \frac{1}{\sqrt{2\pi}} = - \int_0^{\infty} \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx \end{aligned}$$

$$D^2[Y] = E[Y^2] - \underbrace{E^2[Y]}_{=0} = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx =$$

integration by parts: $u(x) = x, v'(x) = x e^{-\frac{x^2}{2}} \Rightarrow u'(x) = 1, v(x) = -e^{-\frac{x^2}{2}}$

$$= \frac{1}{\sqrt{2\pi}} \underbrace{\left[-x \cdot e^{-\frac{x^2}{2}} \right]_{-\infty}^{\infty}}_{=0} - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} -e^{-\frac{x^2}{2}} dx = 1 \quad \square$$

Example: Let $X \sim N(3, 4)$. What is the probability that X is positive?

$$\begin{aligned} P(X > 0) &= P\left(\frac{X-3}{2} > -\frac{3}{2}\right) = 1 - \Phi\left(-\frac{3}{2}\right) = 1 - (1 - \Phi\left(\frac{3}{2}\right)) = \Phi\left(\frac{3}{2}\right) \approx 0.9332 \\ &\quad \begin{array}{c} \nearrow \\ \frac{X-3}{2} \sim N(0, 1) \end{array} \quad \begin{array}{c} \nwarrow \\ \text{prop} \end{array} \end{aligned}$$

II. Stirling's formula

The importance of the normal distribution is shown in the following theorem.

Thm. (DeMoivre-Laplace)

Let X be a random variable with distribution $\text{Bin}(n, p)$.
Then for every $a < b \in \mathbb{R}$.

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \Phi(b) - \Phi(a),$$

where Φ is the distribution function of the standard normal distribution.

Note: Unlike the Poisson approximation, here $0 < p < 1$ is fixed and only n changes.

In the proof of the DeMoivre-Laplace theorem we'll have to handle factorials of big natural numbers. The next theorem will help us out.

Thm. (Stirling's formula)

$$\lim_{n \rightarrow \infty} \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}} = 1, \text{ precisely } \forall n > 0: 1 < \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}} < e^{\frac{1}{12n}}.$$

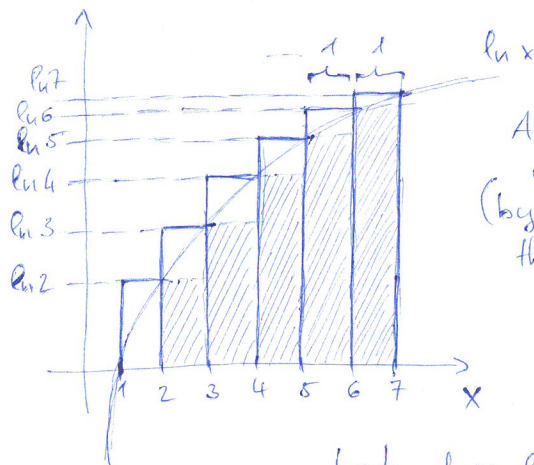
Propo $\exists d \in \mathbb{R}$ such that

$$\forall n > 0, 1 < \frac{n!}{\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}} < e^{\frac{1}{12n}}, \text{ that is } \lim_{n \rightarrow \infty} \frac{n!}{\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}} = 1.$$

Later, using the De Moivre-Laplace theorem, we'll see that $e^d = \sqrt{2\pi}$. Until then, we write e^d in its place as the constant term.

proof:

We want to approximate $n!$. What would be a good candidate?



As the figure implies
(by comparing the areas)

$$\int_1^n \ln x \, dx \leq \ln n! = \sum_{k=1}^n \ln k \leq \int_1^{n+1} \ln x \, dx$$

$$n \ln n - n \leq \ln n! \leq (n+1) \ln(n+1) - n$$

Let's approximate $\ln n!$ with a number within these bounds $\Rightarrow (n + \frac{1}{2}) \ln n - n$.

Let $d_n = \ln n! - [(n + \frac{1}{2}) \ln n - n]$, we are going to show that:

1) d_n is monotonically decreasing, so $\lim_{n \rightarrow \infty} d_n = d \in [-\infty, \infty)$ exists,

2) $\forall n > 0, d < d_n < d + \frac{1}{12n}$

These imply $e^d < e^{d_n} = \frac{n!}{\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}} < e^d e^{\frac{1}{12n}} \Rightarrow 1 < \frac{n!}{\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}} < e^{\frac{1}{12n}}$.

Now we prove 1) and 2).

$$\begin{aligned} 1) \quad d_n - d_{n+1} &= \ln n! - [(n + \frac{1}{2}) \ln n - n] - \ln(n+1)! + [(n + \frac{3}{2}) \ln(n+1) - (n+1)] = \\ &= -(n + \frac{1}{2}) \ln n - \ln(n+1) + (n + \frac{3}{2}) \ln(n+1) - 1 = (n + \frac{1}{2}) \ln \left(\frac{n+1}{n}\right) - 1 = \left(\frac{2n+1}{2}\right) \ln \left(\frac{1 + \frac{1}{2n+1}}{1 - \frac{1}{2n+1}}\right) - 1 = \\ &= \frac{1}{2t} \left[\ln(1+t) - \ln(1-t) \right] - 1 = \frac{1}{2t} \left[-\sum_{k=1}^{\infty} \frac{t^k}{k} + \sum_{k=1}^{\infty} (-1)^k \frac{t^k}{k} \right] - 1 = \end{aligned}$$

$$\ln(1-x) = -\sum_{k=1}^{\infty} \frac{x^k}{k}, \ln(1+x) = -\sum_{k=1}^{\infty} (-1)^k \frac{x^k}{k} \quad \text{Taylor series around 0.}$$

$$\text{even terms cancel out} \Rightarrow \frac{1}{2t} \sum_{l=0}^{\infty} \frac{2t^{2l+1}}{2l+1} - 1 = \sum_{l=0}^{\infty} \frac{t^{2l}}{2l+1} - 1 = \sum_{l=1}^{\infty} \frac{t^{2l}}{2l+1} > 0 \Rightarrow d_n \text{ is monotonically decreasing.}$$

$$2) \quad d_n - d_{n+1} = \sum_{l=1}^{\infty} \frac{t^{2l}}{2l+1} < \frac{1}{3} \sum_{l=0}^{\infty} t^{2l} = \frac{1}{3} \frac{1}{1-t^2} = \frac{1}{3} \frac{\left(\frac{1}{2n+1}\right)^2}{1 - \left(\frac{1}{2n+1}\right)^2} = \frac{1}{3} \frac{1}{(2n+1)^2 - 1} = \frac{1}{12} \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$\Rightarrow \text{For } n < m \in \mathbb{N}, 0 < d_n - d_m \leq \sum_{l=1}^{m-n} d_{n+l-1} - d_{n+l} < \sum_{l=1}^{m-n} \frac{1}{12} \left(\frac{1}{n+l-1} - \frac{1}{n+l}\right) = \frac{1}{12n} - \frac{1}{12m} < \frac{1}{12n}$$

$\Rightarrow (d_n)_{n \geq 0}$ is a Cauchy-sequence, thus $\exists \lim_{n \rightarrow \infty} d_n = d \in \mathbb{R}$.

Moreover, $\forall n > 0, d < d_n < d + \frac{1}{12n}$.

□